

Enterprise Sales Data Warehouse

AWS project implementation and verified results

Final report | October 9, 2026 | Batch proof of concept

SQL Server / Amazon S3 / AWS Glue / Amazon Athena / Amazon Redshift Serverless

COMPLETE | 4 datasets | 21 validated records | 5 reporting views

Executive summary

Built and validated a batch analytics pipeline from Microsoft SQL Server EnterpriseSalesDB through manually exported CSV files, Amazon S3, AWS Glue ETL, and Amazon Redshift Serverless. Amazon Athena was used to validate processed data. Four warehouse tables contain 21 records in total, and five analytical views were created and tested.

Scope boundary: source extraction and upload were manual. AWS Database Migration Service (DMS), change data capture (CDC), automatic ingestion, and a BI dashboard were not implemented.

Implemented architecture

SQL Server (EnterpriseSalesDB) -> manual CSV export -> S3 raw CSV zone -> AWS Glue crawlers and Data Catalog -> AWS Glue transformations -> S3 processed Parquet zone -> Amazon Redshift Serverless -> SQL reporting views.

Amazon Athena validated the processed Parquet datasets before Redshift loading. Redshift COPY read Parquet files from S3 with the REGION parameter in the successful project run.

AWS inventory and configuration

Component	Implemented configuration
S3 bucket	enterprise-sales-data-li us-east-2 (Ohio)
Glue catalog database	enterprise_sales_db
Raw crawler	enterprise-sales-raw-crawler On demand
Processed crawler	enterprise-sales-processed-crawler On demand
Glue role	AWSGlueServiceRole-EnterpriseSales
Redshift namespace	enterprise-sales us-east-1 (N. Virginia)
Redshift workgroup	enterprise-sales-workgroup Available
Redshift database/schema	dev / enterprise_sales
Redshift capacity	4 base RPU's; 4 max RPU's
Redshift IAM role	RedshiftServerlessEnterpriseSalesRole (default)
S3 access policy	RedshiftEnterpriseSalesS3ReadPolicy

Component	Implemented configuration
Budget	EnterpriseSales-AWS-Monthly-Budget \$5/month alerts
Redshift usage limit	10 RPU-hours monthly Turn off user queries

Source data and storage

Dataset	Source rows	Raw CSV	Processed Parquet
Customers	5	raw/customers/customers.csv	processed/customers/
Products	5	raw/products/products.csv	processed/products/
Orders	5	raw/orders/orders.csv	processed/orders/
OrderDetails	6	raw/orderdetails/orderdetails.csv	processed/orderdetails/

All paths are under s3://enterprise-sales-data-li/. CSVs included headers. Glue ETL converted datasets to Parquet and normalized column names to snake_case.

Warehouse tables and data types

Table	Columns
enterprise_sales.customers	customer_id BIGINT; first_name VARCHAR(100); last_name VARCHAR(100); email VARCHAR(255); city VARCHAR(100); state VARCHAR(50); created_date VARCHAR(50)
enterprise_sales.products	product_id BIGINT; product_name VARCHAR(100); category VARCHAR(50); unit_price DOUBLE PRECISION; created_date VARCHAR(50)
enterprise_sales.orders	order_id BIGINT; customer_id BIGINT; order_date VARCHAR(50); order_status VARCHAR(30); total_amount DOUBLE PRECISION
enterprise_sales.orderdetails	order_detail_id BIGINT; order_id BIGINT; product_id BIGINT; quantity BIGINT; unit_price DOUBLE PRECISION

Design note: monetary columns were initially defined as DOUBLE PRECISION, which caused floating-point display artifacts in a report. Reporting queries cast amounts to DECIMAL(18,2). For a production redesign, store money in DECIMAL and dates in DATE/TIMESTAMP rather than VARCHAR.

Data quality and load verification

Dataset	Source / Athena validated	Redshift verified	Result
Customers	5	5	Pass
Products	5	5	Pass
Orders	5	5	Pass

Dataset	Source / Athena validated	Redshift verified	Result
OrderDetails	6	6	Pass
TOTAL	21	21	Pass

Athena raw/processed row-count comparisons passed; duplicate and null checks were reported as passed. Final Redshift UNION ALL COUNT(*) query returned 5, 5, 5, and 6. SQL JOIN between Customers and Orders returned five rows. These counts verify the sample dataset, not production-scale performance.

Verified analytical results

Metric	Observed result
Total orders	5
Total sales revenue	\$2,779.94
Average order value	\$555.98
Monthly sales	September 2026: 5 orders, \$2,779.94
Category: Electronics	4 units, \$2,029.96
Category: Furniture	2 units, \$749.98
Highest-revenue product	Laptop: 2 units, \$1,799.98
Other product sales	Standing Desk \$499.99; Office Chair \$249.99; Monitor \$199.99; Wireless Mouse \$29.99

Customer sales report: John Smith 2 orders / \$1,129.97; Emily Davis 1 / \$899.99; Michael Brown 1 / \$499.99; Sarah Johnson 1 / \$249.99. One customer has no matching order in the inner-join report. Category and product revenues reconcile to \$2,779.94.

Daily sales reporting results

Sales date	Orders	Revenue
2026-09-20	1	\$929.98
2026-09-21	1	\$249.99
2026-09-22	1	\$499.99
2026-09-23	1	\$199.99
2026-09-24	1	\$899.99

These daily figures were provided as expected query results in the project walkthrough; the user confirmed that the daily-sales query succeeded, but the individual daily output rows were not separately transcribed from a screenshot. Treat the date-by-date values as walkthrough expectations rather than independently verified row-level evidence.

Analytical reporting views

View	Purpose / validation
vw_monthly_sales	Monthly order count and revenue; created and tested
vw_customer_sales	Orders and revenue per customer; created and tested
vw_category_sales	Units and revenue by product category; created and tested
vw_daily_sales	Daily order count and revenue; created and tested
vw_product_performance	Product-level units and revenue; screenshot confirmed five products

A proposed sixth view (vw_customer_order_history) was not created; work shifted to finalizing the project.

Cost and operational controls

Item	Observed amount or configuration
AWS Glue interactive sessions	6.31 DPU-hours / \$2.78
AWS Glue ETL jobs	1.025 DPU-hours / \$0.45
AWS Glue crawlers	0.765 DPU-hours / \$0.34
AWS Glue gross usage	\$3.57
AWS Glue credit/adjustment	-\$3.57
Billing estimate at review	\$0.00 amount due (pending)
Redshift trial balance (last screenshot)	\$299.15 remaining; expiry shown Jan 5, 2027
AWS budget	\$5/month with 85%, 100%, and forecasted 100% alerts
Redshift usage limit	10 RPU-hours per month; turn off user queries

Eight Glue interactive sessions were observed with Timeout status; no running sessions were shown. Four Glue ETL jobs had no schedules, Glue triggers list was empty, and both crawlers were On demand. Billing and credits can change after the screenshots; budget alerts and RPU limits do not guarantee a zero bill or stop all service charges.

Security, limitations and production recommendations

IAM permissions were granted to Redshift for listing the project S3 bucket and reading processed objects. Avoid publishing account IDs, access keys, credentials, private bucket contents, or unredacted console screenshots in a public repository.

The S3 bucket and Redshift workgroup were in different regions (Ohio and N. Virginia), so cross-region access was part of this demo. For production, assess data transfer, performance, and region alignment.

The pipeline was a manual batch proof of concept. It has no demonstrated incremental ingestion, CDC, orchestration, automated test suite, failure retry, monitoring dashboard, or production-scale benchmarks. Redshift informational primary keys do not enforce uniqueness; enforce data quality upstream.

Suggested Phase 2: automate source ingestion; add orchestration and alerts; adopt DECIMAL/DATE types; introduce idempotent incremental loads; add version-controlled ETL code and CI/CD; optionally build a BI dashboard.

Final project acceptance

Acceptance check	Outcome
Source data exported and uploaded	Completed
Four Glue ETL transformations	Completed
Processed data cataloged and validated	Completed
Redshift tables created and populated	Completed
21 source records reconciled to warehouse	Passed
Five reusable reporting views	Created and tested
Budget and Redshift usage limit	Configured
AWS DMS / CDC	Not in scope
GitHub portfolio	Not yet published

Conclusion: The implemented batch warehouse proof of concept is complete. GitHub publication is a separate portfolio task.